

Shohaib Ibne Monju

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Phone No: +8801795709898

Education

Bachelor of Science in Materials and Metallurgical Engineering

Feb 2020 – Mar 2025

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

Relevant Courses: Electrical & Magnetic Properties of Materials, Phase Diagrams, Mechanical Behavior of Materials, Materials Processing for Micro & Nano-Systems, Computer Programming.

GPA: 3.18/4.00

Research Interests

- Machine Learning
- Density Functional Theory
- Energy Conversion
- Molecular Dynamics
- Energy Storage Devices
- Electrochemistry

Research Experience

Undergraduate Thesis

Effect of Cryogenic Deformation on Properties of AA6063 Alloys

Nov 2023 – Mar 2025

Investigated the influence of deep cryogenic treatment and subsequent rolling on the microstructure and mechanical behavior of AA6063 aluminum alloy.

- Performed liquid nitrogen immersion followed by rolling at varying reductions to assess deformation effects.
- Characterized microstructural evolution using optical and scanning electron microscopy to evaluate grain refinement and intermetallic dispersion.
- Found enhanced hardness and strength due to suppressed recovery and refined microstructure, though accompanied by reduced ductility and impact toughness.

Supervisor: [Dr. A. K. M. Bazlur Rashid](#)

Undergraduate Researcher (Remote)

PI & Group Lead: [Dr. Saquib Ahmed](#), SUNY – Buffalo State University, New York, USA

Jan 2023 – Present

[Ahmed Research Group](#)

Work Focus:

- Integrated machine-learning-based prediction models with materials datasets for structure–property analysis.
- Performed density functional theory (DFT) calculations to validate phase stability of high-entropy alloys.
- Conducted molecular dynamics (MD) simulations to study defect behavior and mechanical responses in multi-principal element alloys.

Tools & Methods: Python (pandas, scikit-learn, matplotlib), VASP (DFT), LAMMPS (MD), VESTA, OVITO, Atomsk, Packmol, MATLAB, OriginLab.

Publications

[1] Abdul Hamid Rumman, Kaushik Barua, **Shohaib Ibne Monju**, Mohd Rakibul Hasan Abed, Sadika Jannath Tan-Ema, Jafar F. Al-Sharab, Saquib Ahmed;

“Classification of CoCr-based magnetic thin films via GLCM texture features extracted from EFTEM images and machine learning,” *AIP Advances*, vol. 14, no. 11, p. 115017, **Nov. 2024**. [DOI](#)

[2] Miah Abdullah Sahriar, Abdul Hamid Rumman, Ahammad Ullah, Kaushik Barua, **Shohaib Ibne Monju**, Munim Shahriar Jawad, Md. Atik Faisal, Ridwan Radit Ahsan, Houk Jang, Saquib Ahmed;

“Optical image analysis of WSe₂ thresholding for layer detection,” *Computational Materials Science*, vol. 253, p. 113888, **May 2025**. [DOI](#)

[3] Abdul Hamid Rumman, Kaushik Barua, **Shohaib Ibne Monju**, Aabrar Bin Salahuddin, Md. Tohidul Islam, Saquib Ahmed;
“Accelerated Design of High-Entropy Alloys via Probabilistic Substitution and Machine Learning with First-Principles Validation of Phase Stability,” *Journal of Alloys and Compounds*, vol. 1048, p. 185217, Dec. 2025. [DOI](#)

Capstone Project

Synthesizing Decaffeinating Filter from Cellulose Powder to Control Caffeine in Coffee Dec. 2022 – Sep. 2023

- Designed and synthesized a cellulose–tea composite filter using EGDMA crosslinker prepared from ethylene glycol and acrylic acid.
- Optimized pulp composition and pressing conditions to achieve caffeine removal efficiency.

Supervisor: [Dr. Ahmed Sharif](#)

Industrial Attachment

Industrial Training

June 2024

Had the opportunity to visit DBL, a leading ceramics manufacturing company in Bangladesh.

- Observed raw materials and machinery for wall/floor tile manufacturing.
- Studied effects of raw materials on Modulus of Rupture (MOR).
- Identified common manufacturing flaws causing fractures and methods to mitigate them.

Technical Skills

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- **Experimental Skills:** Uniaxial Pressing, Sand Molding, Metallography, Microstructure Analysis, Chemical Analysis, Electrodeposition, Corrosion Testing.
 - **Software & Computational Skills:** VASP (DFT), OriginLab, CES Edupack, MATLAB, SolidWorks, MS Office, LaTeX.
Molecular Dynamics: Basic simulation via LAMMPS; structure generation / visualization with VESTA, OVITO, Atomsk, Packmol.
 - **Programming Languages:** C, C++, Python (matplotlib, pandas, scikit-learn).

Test Scores

TOEFL: 110/120 (R: 30, L: 30, S: 23, W: 27)

GRE: 311 (Q: 164, V: 147, AWA: 3.5)

Extra-Curricular Activities

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- Vice President, BUET Career Club (2023–2024)
 - Publication Secretary, SAMME (Students' Association of MME)
 - Part-time Script Evaluator, Udvash Academic & Admission Care
 - Instructor, ScholaraiD Academic & Admission Aid

References

Dr. Saquib Ahmed

Associate Professor, Engineering Technology
The State University of New York, Buffalo State University

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[Google Scholar](#) | [Homepage](#)

Dr. Kazi Md. Shorowordi

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[Google Scholar](#) | [Department Profile](#)